AERO70015/96012 Mathematics 3 Problem Set #4

Problem Sheet 4

4.1 PR4_1

(a) Using the convolution theorem, show that

$$\mathcal{L}^{-1}\left\{\frac{1}{s\,(s+a)}\right\} = \frac{1}{a}\,\left(1-e^{-a\,t}\right).$$

(b) Hence obtain the solution of the o.d.e.

$$x''(t) + a\,x'(t) = f(t), \quad x(0) = x_0, \quad x'(0) = x_1,$$

where $f(t)$ is an arbitrary function.

Show that the steady-state value of x is given by

$$x(\infty) = \frac{1}{a}\int_0^\infty f(u)\,du + x_0 + \frac{x_1}{a}.$$

(a) $L^{-1}\left\{\frac{1}{s(s+a)}\right\} = L^{-1}\left\{\frac{1}{s}\right\} L^{-1}\left\{\frac{1}{s+a}\right\}$

$= \int_0^t e^{-a\tau}\cdot 1\, d\tau$

$= \left[-\frac{1}{a}e^{-a\tau}\right]_0^t$

$= \frac{1}{a}(1-e^{-at})$

(b) Apply Laplace transformations:

$s^2X - sx(0) - x'(0) + a[sX - x(0)] = F(s)$

$\therefore (s^2+as)X - (a+s)x_0 - x_1 = F(s)$

$\therefore X = \frac{1}{s(s+a)}[F(s) + (s+a)x_0 + x_1]$

$\therefore x = L^{-1}\left\{\frac{F(s)}{s(s+a)}\right\} + L^{-1}\left\{\frac{s+a}{s(s+a)}x_0\right\} + L^{-1}\left\{\frac{1}{s(s+a)}x_1\right\}$

$= L^{-1}\left\{\frac{1}{s(s+a)}\right\} L^{-1}\{F(s)\} + L^{-1}\left\{\frac{x_0}{s+a}\right\} + L^{-1}\left\{\frac{a}{s(s+a)}x_0\right\} + L^{-1}\left\{\frac{1}{s(s+a)}x_1\right\}$

$= \frac{1}{a}\int_0^t (1-e^{-a(t-u)}) f(u)\,du + e^{-at}x_0 + (1-e^{-at})x_0 + \frac{1}{a}(1-e^{-at})x_1$

$\therefore x(\infty) = \frac{1}{a}\int_0^\infty f(u)\,du + x_0 + \frac{1}{a}x_1$

4.2 PR4_2

(a) Using the convolution theorem, show that if

$$y(x) = f(x) + \int_0^x g(x-u)\ y(u)\ du,$$

then $Y(s) = F(s)/(1 - G(s))$, where Y, F and G are the Laplace transforms of y, f and g respectively.

(b) Hence solve the integral equation

$$y(x) = \sin(3\ x) + \int_0^x \sin(x-u)\ y(u)\ du.$$

(a) Apply Laplace transformations:

$$Y(s) = F(s) + G(s)Y(s)$$

$$\therefore [1 - G(s)]Y(s) = F(s)$$

$$\therefore Y(s) = \frac{F(s)}{1-G(s)}$$

(b) $$Y(s) = \frac{F(s)}{1-G(s)} = \frac{\frac{3}{s^2+9}}{1-\frac{1}{s^2+1}}$$

$$= \frac{3(s^2+1)}{(s^2+9)(s^2+1-1)}$$

$$\therefore\ Y(s) = \frac{3(s^2+1)}{s^2(s^2+9)} = \frac{\frac{1}{3}}{s^2} + \frac{\frac{8}{3}}{s^2+9}$$

$$\therefore\ y(x) = \mathcal{L}^{-1}\{Y(s)\} = \frac{1}{3}\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} + \frac{8}{9}\mathcal{L}^{-1}\left\{\frac{3}{s^2+9}\right\}$$

$$= \frac{1}{3}x + \frac{8}{9}\sin(3x)$$

4.3 PR4_3

Find

$$\mathcal{L}^{-1}\left\{\frac{s}{(s^2+\omega^2)^2}\right\}$$

(with ω a positive constant) by the following methods:

(a) Splitting into partial fractions and using the shift theorems and known transforms;

(b) Using the convolution theorem together with known transforms;

(c) Using the complex inversion formula.

(a) $$\frac{s}{(s^2+\omega^2)^2} = \frac{1}{4i\omega}\left[-\frac{1}{(s+i\omega)^2} + \frac{1}{(s-i\omega)^2}\right]$$

$$\therefore\ \mathcal{L}^{-1}\left\{\frac{s}{(s^2+\omega^2)^2}\right\} = \frac{1}{4i\omega}\left\{-\frac{1}{(s+i\omega)^2}\right\} + \frac{1}{4i\omega}\left\{\frac{1}{(s-i\omega)^2}\right\}$$

$$= \frac{1}{4i\omega}(-e^{-i\omega t}t + e^{i\omega t}t) = \frac{2i\sin(\omega t)t}{4i\omega} = \frac{\sin(\omega t)t}{2\omega}$$

(b) $\dfrac{s}{(s^2+\omega^2)^2} = \dfrac{s}{(s^2+\omega^2)}\dfrac{1}{(s^2+\omega^2)} = \dfrac{1}{\omega}\dfrac{s}{(s^2+\omega^2)}\dfrac{\omega}{(s^2+\omega^2)}$

$\therefore L^{-1}\left\{\dfrac{s}{(s^2+\omega^2)^2}\right\} = \dfrac{1}{\omega}\int_0^t \cos(\omega u)\sin[\omega(t-u)]\,du$

where $\sin a\cos b - \cos a\sin b = \sin(a-b)$

$$\therefore L^{-1}\left\{\frac{s}{(s^2+\omega^2)^2}\right\} = \frac{1}{\omega}\int_0^t \cos(\omega u)[\sin(\omega t)\cos(\omega u) - \cos(\omega t)\sin(\omega u)]\,du$$

$$= \frac{\sin(\omega t)}{\omega}\int_0^t \cos^2(\omega u)\,du - \frac{\cos(\omega t)}{\omega}\int_0^t \cos(\omega u)\sin(\omega u)\,du$$

$$= \frac{\sin(\omega t)}{\omega}\int_0^t \frac{1}{2}[1+\cos(2\omega u)]\,du - \frac{\cos(\omega t)}{\omega}\int_0^t \frac{1}{2}\sin(2\omega u)\,du$$

$$= \frac{1}{\omega}\sin(\omega t)\left[\frac{1}{2}u + \frac{1}{4\omega}\sin(2\omega u)\right]_0^t - \frac{1}{\omega}\cos(\omega t)\left[-\frac{1}{4\omega}\cos(2\omega u)\right]_0^t$$

$$= \frac{1}{2\omega}\sin(\omega t)t + \frac{1}{4\omega^2}\sin(2\omega t)\sin(\omega t) + \frac{1}{4\omega^2}[\cos(2\omega t)-1]\cos(\omega t)$$

$$= \frac{1}{2\omega}\sin(\omega t)t + \frac{1}{2\omega}\sin^2(\omega t)\cos(\omega t) + \frac{1}{2\omega^2}[\cos^2(\omega t)-1]\cos(\omega t)$$

$$= \frac{1}{2\omega}\sin(\omega t)t + \frac{1}{2\omega}[\sin^2(\omega t)+\cos^2(\omega t)-1]\cos(\omega t)$$

$$= \frac{\sin(\omega t)t}{2\omega}$$

(c) Let $f(s)e^{st} = \dfrac{se^{st}}{(s^2+\omega^2)^2} = \dfrac{se^{st}}{(s^2+\omega^2)(s^2+\omega^2)} = \dfrac{se^{st}}{(s+i\omega)^2(s-i\omega)^2}$

Let $p = se^{st}$, $q = (s^2+\omega^2)^2$

$\therefore p' = (1+st)e^{st}$, $q' = 4s(s^2+\omega^2)$, $q'' = 12s^2+4\omega^2$, $q''' = 24s$

At $s_0 = i\omega$, $p(s_0) = i\omega e^{i\omega t}$, $p'(s_0) = (1+i\omega t)e^{i\omega t}$, $q(s_0)=0$, $q'(s_0)=0$, $q''(s_0) = -8\omega^2$, $q'''(s_0) = 24i\omega$

At $s_0 = -i\omega$, $p(s_0) = -i\omega e^{-i\omega t}$, $p'(s_0) = (1-i\omega t)e^{-i\omega t}$, $q(s_0)=0$, $q'(s_0)=0$, $q''(s_0) = -8\omega^2$, $q'''(s_0) = -24i\omega$

$$\therefore L^{-1}\left\{\frac{s}{(s^2+\omega^2)^2}\right\} = \operatorname*{Res}_{\omega_0=i\omega} f(z)e^{st} + \operatorname*{Res}_{\omega_0=-i\omega} f(z)e^{st}$$

$$= \frac{2(1+i\omega t)e^{i\omega t}}{-8\omega^2} - \frac{2i\omega e^{i\omega t}(24i\omega)}{3(64\omega^4)} + \frac{2(1-i\omega t)e^{-i\omega t}}{-8\omega^2} - \frac{2i\omega e^{-i\omega t}(24i\omega)}{3(64\omega^4)}$$

$$= -\frac{e^{i\omega t}+e^{-i\omega t}}{4\omega^2} - \frac{it(e^{i\omega t}-e^{-i\omega t})}{4\omega} + \frac{e^{i\omega t}+e^{-i\omega t}}{4\omega^2} = \frac{\sin(\omega t)t}{2\omega}$$

4.4 PR4_4

Solve the partial differential equation

$$\frac{\partial u}{\partial t} + x\,\frac{\partial u}{\partial x} = x, \quad u(x,\ 0) = 0, \quad u(0,\ t) = 0$$

for $u(x,\ t)$ (with $x > 0$, $t > 0$) by taking the Laplace transform with respect to t.

Apply Laplace transform.

$$sU - u(x,0) + x\frac{\partial U}{\partial x} = \frac{x}{s}$$

$$\therefore \frac{\partial U}{\partial x} + \frac{s}{x}U = \frac{1}{s}$$

let integral factor $I(x) = e^{\int \frac{s}{x}dx} = e^{s\ln x} = x^s$

$$\therefore \frac{d}{dx}(Ux^s) = \frac{x^s}{s}$$

$$\therefore Ux^s = \int \frac{x^s}{s}dx = \frac{1}{s(s+1)}x^{s+1} + C$$

$$\therefore U = \frac{x}{s(s+1)} + Cx^{-s} = \frac{x}{s} - \frac{x}{s+1} + Cx^{-s}$$

$$\therefore u = \mathcal{L}^{-1}\{U\} = x(1-e^{-t}) + \mathcal{L}^{-1}\{Cx^{-s}\}$$

$$\because u(0,t) = 0 + \mathcal{L}^{-1}\{Cx^{-s}\} = 0$$

$$\therefore \mathcal{L}^{-1}\{Cx^{-s}\} = 0$$

$$\therefore u = x(1-e^{-t})$$

4.5 PR4_5

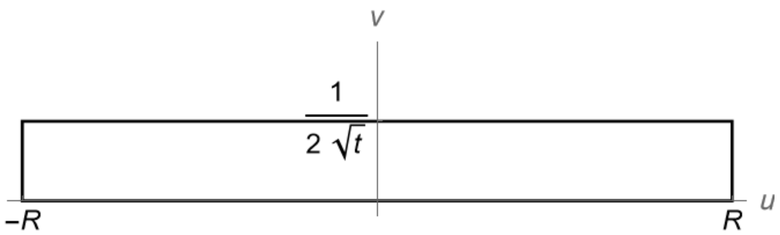
(a) Using a keyhole Bromwich contour, with a branch cut along the negative real axis, show that the inverse Laplace transform of $F(s) = e^{-\sqrt{s}}$ is equal to

$$\frac{1}{\pi} \int_0^{\infty} e^{-r\,t} \sin(\sqrt{r})\, dr.$$

(You may assume that the integrals along the outer curves tend to zero as $R \to \infty$.)

(b) Using the substitution $r\,t = u^2$, show that this is equal in turn to the imaginary part of

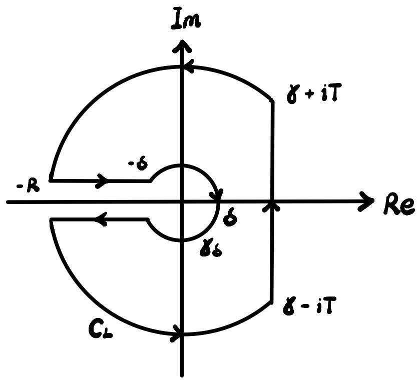
$$\frac{1}{\pi\, t} \int_{-\infty}^{\infty} u\, e^{-u^2}\, e^{i\,u/\sqrt{t}}\, du.$$



(c)

By considering the integral of $w\, e^{-(w-i/[2\sqrt{t}])^2}$ around the contour shown in the figure, where $w = u + i\,v$, and letting R tend to infinity, express this imaginary part in terms of t, π and the known integral $\int_{-\infty}^{\infty} e^{-u^2}\, du = \sqrt{\pi}$, and hence find the inverse Laplace transform of $F(s) = e^{-\sqrt{s}}$.

(a)



$$\oint_c e^{-\sqrt{s}} e^{sz}\, dz$$

$$= -\oint_{\gamma_\delta} e^{-\sqrt{s}} e^{sz}\, dz + \int_R^{\delta} e^{-\sqrt{x} e^{i\frac{\pi}{2}}} e^{sxe^{i\pi}} (e^{i\pi})\, dx + \oint_{C_L} e^{-\sqrt{s}} e^{sz}\, dz$$

$$+ \int_{\delta}^{R} e^{-\sqrt{x} e^{i(-\frac{\pi}{2})}} e^{sxe^{i(-\pi)}} (e^{-i\pi})\, dx + \int_{\gamma - iT}^{\gamma + iT} e^{-\sqrt{s}} e^{sz}\, dz$$

$$\therefore \oint_C e^{-\sqrt{z}} e^{sz}\,dz = -\oint_{\gamma_\delta} e^{-\sqrt{z}} e^{sz}\,dz + \int_\delta^R e^{-i\sqrt{x}} e^{-sx}\,dx + \oint_{C_L} e^{-\sqrt{z}} e^{sz}\,dz$$

$$+ \int_\delta^R -e^{i\sqrt{x}} e^{-sx}\,dx + \int_{\gamma-iT}^{\gamma+iT} e^{-\sqrt{z}} e^{sz}\,dz$$

for $R \to \infty$, assume $\oint_{C_L} e^{-\sqrt{z}} e^{sz}\,dz \to 0$

for $\delta \to 0$,

$$\oint_{\gamma_\delta} e^{-\sqrt{z}} e^{sz}\,dz = \int_0^{2\pi} e^{-\sqrt{\delta}e^{i\frac{\theta}{2}}} e^{s\delta e^{i\theta}} (i\delta e^{i\theta})\,d\theta \longrightarrow 0$$

for closed contour of analytic function, $\oint_C e^{-\sqrt{z}} e^{sz}\,dz = 0$

Therefore,

$$\int_0^\infty e^{-i\sqrt{x}} e^{-sx}\,dx + \int_0^\infty -e^{i\sqrt{x}} e^{-sx}\,dx + \int_{\gamma-i\infty}^{\gamma+i\infty} e^{-\sqrt{z}} e^{sz}\,dz = 0$$

$$\therefore L^{-1}\{e^{-\sqrt{s}}\} = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} e^{-\sqrt{z}} e^{sz}\,dz$$

$$= -\frac{1}{2\pi i} \int_0^\infty (e^{-i\sqrt{x}} - e^{i\sqrt{x}}) e^{-sx}\,dx$$

$$= -\frac{1}{2\pi i} \int_0^\infty -2i\sin\sqrt{x}\, e^{-sx}\,dx$$

$$= \frac{1}{\pi} \int_0^\infty \sin\sqrt{x}\, e^{-sx}\,dx$$

Let $x = r$, $s = t$, therefore $L^{-1}\{e^{-\sqrt{s}}\} = \frac{1}{\pi}\int_0^\infty \sin\sqrt{r}\, e^{-rt}\,dr$

(b) Let $rt = u^2$, therefore $\frac{dr}{du} = \frac{2u}{t}$

$$\frac{1}{\pi}\int_0^\infty \sin\sqrt{r}\, e^{-rt}\,dr = \frac{1}{\pi}\int_0^\infty \sin\left(\frac{u}{\sqrt{t}}\right) e^{-u^2} \frac{2u}{t}\,du$$

$$= \text{Im}\left\{ \frac{1}{\pi t} \int_0^\infty u e^{-u^2} e^{i\frac{u}{\sqrt{t}}}\,du \right\}$$

(c) $$\oint_C w e^{-(w-\frac{i}{2\sqrt{t}})^2}\,dw = \int_{-R}^{R} u e^{-(u-\frac{i}{2\sqrt{t}})^2}\,du + \int_0^{\frac{1}{2\sqrt{t}}} (R+iv) e^{-[R+i(v-\frac{1}{2\sqrt{t}})]^2}\,dv$$

$$+ \int_R^{-R} (u + i\frac{1}{2\sqrt{t}}) e^{-u^2}\,du + \int_{\frac{1}{2\sqrt{t}}}^0 (-R+iv) e^{-[-R+i(v-\frac{1}{2\sqrt{t}})]^2}\,dv$$

for $R \to \infty$, $\int_0^{\frac{1}{2\sqrt{t}}} (R+iv)e^{-[R+i(v-\frac{1}{2\sqrt{t}})]^2}\,dv \longrightarrow 0$

$$\int_{\frac{1}{2\sqrt{t}}}^{0} (-R+iv)e^{-[-R+i(v-\frac{1}{2\sqrt{t}})]^2}\,dv \longrightarrow 0$$

$$\therefore \oint_c we^{-(w-\frac{i}{2\sqrt{t}})^2}dw = e^{\frac{1}{4t}}\int_{-\infty}^{\infty} ue^{-u^2}e^{i\frac{u}{\sqrt{t}}}\,du + \int_{\infty}^{-\infty}(ue^{-u^2}+\frac{1}{2\sqrt{t}}ie^{-u^2})\,du$$

Along the closed contour with analytic function, $\oint_c we^{-(w-\frac{i}{2\sqrt{t}})^2}dw = 0$

Therefore,

$$e^{\frac{1}{4t}}\int_{-\infty}^{\infty} ue^{-u^2}e^{i\frac{u}{\sqrt{t}}}\,du = \int_{-\infty}^{\infty}(ue^{-u^2}+\frac{1}{2\sqrt{t}}ie^{-u^2})\,du$$

$$= \left[-\tfrac{1}{2}e^{-u^2}\right]_{-\infty}^{\infty} + \frac{1}{2\sqrt{t}}i\int_{-\infty}^{\infty} e^{-u^2}du$$

$$\therefore \mathrm{Im}\left\{\int_{-\infty}^{\infty} ue^{-u^2}e^{i\frac{u}{\sqrt{t}}}\,du\right\} = \frac{e^{-\frac{1}{4t}}}{2\sqrt{t}}\sqrt{\pi}$$

$$\therefore \mathrm{Im}\left\{\frac{1}{\pi t}\int_{-\infty}^{\infty} ue^{-u^2}e^{i\frac{u}{\sqrt{t}}}\,du\right\} = \frac{e^{-\frac{1}{4t}}}{2\sqrt{\pi}\,t^{\frac{3}{2}}}$$

$$\therefore L^{-1}\{e^{-\sqrt{s}}\} = \frac{1}{\pi}\int_0^{\infty}\sin\sqrt{r}\,e^{-rt}\,dr = \frac{e^{-\frac{1}{4t}}}{2\sqrt{\pi}\,t^{\frac{3}{2}}}$$

4.6 PR4_6

(a) By using the residue theorem and the complex inversion formula for the Laplace transform, prove that

$$\mathcal{L}^{-1}\left\{\frac{1}{s\,(s^2+2\,s+3)}\right\} = \frac{1}{3}\left(1 - e^{-t}\,\cos(\sqrt{2}\,t) - \frac{e^{-t}}{\sqrt{2}}\,\sin(\sqrt{2}\,t)\right).$$

(b) By taking a Laplace transform and using the result from part (a), solve

$$\frac{\partial^2 u}{\partial t^2} + 2\,\frac{\partial u}{\partial t} + x\,\frac{\partial u}{\partial x} = x^3, \quad u(x,\,0) = u_t(x,\,0) = u(0,\,t) = 0.$$

(a) let $f(s) = \dfrac{e^{st}}{s(s^2+2s+3)}$

for $s(s^2+2s+3)=0$, $s_0 = 0$ or $-1 \pm \sqrt{2}i$

let $p = e^{st}$, $q = s(s^2+2s+3)$

$\therefore q' = (s^2+2s+3) + s(2s+2)$

$\therefore \operatorname*{Res}_{s_0=0} f(z) = \frac{1}{3}$

$$\operatorname*{Res}_{s_0=-1+\sqrt{2}i} f(z) = \frac{e^{-t}e^{\sqrt{2}it}}{(-1+\sqrt{2}i)2\sqrt{2}i} = \frac{e^{-t}e^{\sqrt{2}it}}{-4-2\sqrt{2}i}$$

$$\operatorname*{Res}_{s_0=-1-\sqrt{2}i} f(z) = \frac{e^{-t}e^{-\sqrt{2}it}}{(-1-\sqrt{2}i)(-2\sqrt{2}i)} = \frac{e^{-t}e^{-\sqrt{2}it}}{-4+2\sqrt{2}i}$$

$$\therefore \mathcal{L}^{-1}\left\{\frac{1}{s(s^2+2s+3)}\right\} = \frac{1}{3} - \frac{e^{-t}e^{\sqrt{2}it}}{4+2\sqrt{2}i} - \frac{e^{-t}e^{-\sqrt{2}it}}{4-2\sqrt{2}i}$$

$$= \frac{1}{3} - \frac{(4-2\sqrt{2}i)e^{-t}e^{\sqrt{2}it}}{24} - \frac{(4+2\sqrt{2}i)e^{-t}e^{-\sqrt{2}it}}{24}$$

$$= \frac{1}{3} - \frac{e^{-t}(e^{\sqrt{2}it}+e^{-\sqrt{2}it})}{6} + \frac{\sqrt{2}ie^{-t}(e^{\sqrt{2}it}+e^{-\sqrt{2}it})}{12}$$

$$= \frac{1}{3} - \frac{e^{-t}\cos(\sqrt{2}t)}{3} - \frac{e^{-t}\sin(\sqrt{2}t)}{3\sqrt{2}}$$

$$= \frac{1}{3}\left[1 - e^{-t}\cos(\sqrt{2}t) - \frac{e^{-t}}{\sqrt{2}}\sin(\sqrt{2}t)\right]$$

(b) $s^2U - su(x,0) - u_t(x,0) + 2sU - u(x,0) + x\frac{\partial U}{\partial x} = \frac{x^3}{s}$

$$\therefore \frac{\partial U}{\partial x} + \frac{s^2+2s}{x}U = \frac{x^2}{s}$$

Let $I(x) = e^{\int \frac{s^2+2s}{x}dx} = x^{s^2+2s}$

$$\therefore \frac{d}{dx}(Ux^{s^2+2s}) = \frac{1}{s}x^{s^2+2s+2}$$

$\therefore Ux^{s^2+2s} = \frac{1}{s(s^2+2s+3)}x^{s^2+2s+3} + C$ $\quad\therefore U = \frac{1}{s(s^2+2s+3)}x^3 + Cx^{-s^2-2s}$

$$\therefore u = \frac{x^3}{3}\left[1 - e^{-t}\cos(\sqrt{2}t) - \frac{e^{-t}}{\sqrt{2}}\sin(\sqrt{2}t)\right] + \mathcal{L}^{-1}\{Cx^{-s^2-2s}\}$$

where $\mathcal{L}^{-1}\{Cx^{-s^2-2s}\} = 0$ as $u(0,t) = 0$

4.7 PR4_7

(a) Show that if $f(t) = \int_t^\infty \frac{g(u)}{u}\, du$ then $F(s) = \frac{1}{s}\int_0^s G(q)\, dq$.

(b) Show that the Laplace transform of $f(t) = \int_t^\infty \frac{e^{-u}}{u}\, du$ is $\frac{1}{s}\ln(s+1)$.

(a) $F(s) = \int_0^\infty e^{-st} \int_t^\infty \frac{g(u)}{u}\, du\, dt$

$$= \int_0^\infty e^{-st} \int_0^u \frac{g(u)}{u}\, dt\, du$$

$$= \int_0^\infty \frac{g(u)}{u} \int_0^u e^{-st}\, dt\, du$$

$$= \int_0^\infty \left(-\frac{1}{s}e^{-ut} + \frac{1}{s}\right) \frac{g(u)}{u}\, du$$

$$= \frac{1}{s}\int_0^\infty (1 - e^{-tu}) \frac{g(u)}{u}\, du$$

$\because \int_0^s G(q)\, dq = \int_0^s \int_0^\infty e^{-qu} g(u)\, du\, dq$

$$= \int_0^\infty \int_0^s e^{-qu} g(u)\, dq\, du$$

$$= \int_0^\infty g(u) \int_0^s e^{-qu}\, dq\, du$$

$$= \int_0^\infty (1 - e^{-su}) \frac{g(u)}{u}\, du$$

$\therefore F(s) = \frac{1}{s}\int_0^s G(q)\, dq$

(b) $F(s) = \frac{1}{s}\int_0^s G(q)\, dq$

where $g(u) = e^{-u}$

$\therefore G(q) = \frac{1}{q+1}$

$\therefore F(s) = \frac{1}{s}\int_0^s G(q)\, dq = \frac{1}{s}\int_0^s \frac{1}{q+1}\, dq$

$$\therefore F(s) = \frac{1}{s}\ln(s+1)$$